

Diagnostics: Can This Model Be Trusted?

Module 4, Part 2 (November 13)

QBA 5330

Recall From Earlier

We trusted every t-test and confidence interval at face value.

This part: is that trust earned?

The LINE Assumptions

- Linearity
- Independence
- Normality
- Equal variance

A Healthy Model

```
1 model <- lm(sales ~ tv + radio + online, data = ads)
2 ols_test_breusch_pagan(model)
```

Breusch Pagan Test for Heteroskedasticity

Ho: the variance is constant

Ha: the variance is not constant

Data

Response : sales

Variables: fitted values of sales

Test Summary

DF = 1

A Broken Model

```
1 model_het <- lm(y3 ~ x3)
2 ols_test_breusch_pagan(model_het)
```

Breusch Pagan Test for Heteroskedasticity

Ho: the variance is constant

Ha: the variance is not constant

Data

Response : y3

Variables: fitted values of y3

Test Summary

DF = 1

Same test. Same code. Opposite conclusion.

Activity: Diagnose the Model

Four plots. For each: which LINE assumption (if any) is violated, and what do you do next?

- Plot A: patternless cloud
- Plot B: U-shape
- Plot C: funnel/cone shape
- Plot D: one huge bar in a Cook's distance chart

Debrief

- Plot A → no violation, trust it
- Plot B → linearity, needs a new functional form
- Plot C → equal variance, standard errors unreliable
- Plot D → **not a LINE issue** → one influential point, investigate it

Before the Break

- Every model today predicted a **number**
- What about yes/no outcomes: will this customer churn?
- **Next: logistic regression, live.**